

Week 2 Day 1

Test Case

It is a set of actions executed to verify a particular feature or functionality of your software application.

- Pre-conditions :- The conditions that must be true before the test can run.
- Inputs / test steps :- The data / actions applied to the test object
- Expected result :- what a system should produce
- Post condition / ~~Act~~ Actual Result :- The state of the system or env after the test executes.

Positive test case

These test cases checks the system works correctly when given a valid input

Negative test case

It checks that the system correctly handles invalid inputs, reject them and give an appropriate errors.

Defect density
Defect lifecycle
Test Data

STLC

QA

effective
test cases
how to write

Tal

Pre fix

Versioning

Prioritisation

Test case prioritisation is the deciding the order in which test case should be run.

- Risk based - It test the riskiest Features first. for eg. login, money transfer.
- Coverage based - It picks the test case with highest coverage at first before moving to the next. like - a test case covers 60 line of code at a function having 100 line of code - 60 line cover test case will be run, then 30, the 10.
- Requirement based - Requirement are priorities by stake holder or senior members then the test case are tested corresponding to it. for eg. Stakeholder priority ranking for a hospital management system:
 - i) Patient record retrieval (Priority. High)
 - ii) Appointment reminder via SMS (Priority. Medium)
 - iii) Theme mode toggle (Priority. Low)

Traceability to Requirements

It means keeping a clear link between req. and everything built to test them - test conditions, test cases and defects.

Software Testing life cycle (STLC)

It's a structured process that defines the steps involved in testing a software product

- It provide a systematic approach to a plan, execute and manage testing.
- It help to detect early defects, while improving the quality of software
- It ensure that test coverage and proper reporting for ~~error~~ results.

- Requirement analysis
- Test planning
- Test case development
- Test environment setup
- Test execution
- Test closure.

● Bug is a flaw, defect, error, or fault in a software

Stage of Bug life cycle

- New
- Open
- Fixed
- Retest
- closed
- Rejected

- Deferred - The bug is acknowledged but postponed to a future release due to low priority or other constraints
- Reopen

Bug Report Key Elements

- Bug ID
- Bug Title
- Description
- Severity -

Indicates the impact of the defect on the system (Critical, Major, Minor)

- Priority:

Indicates the urgency with which the defect should be fixed (High, Medium, Low)

- Steps to Reproduce

- Reported By / Reported On

- Status

- Fixed By / Closed On

Effective way to write a test case

- Clarity (write for reader, not yourself)
- one test case - one clear purpose
- cover the full scope but don't go beyond it
- Make it traceable
- Make it repeatable & precise
- keep it independent (if possible)